

## Background information on the metrics

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This special session explores the synergies and challenges of three approaches. As basic background information, Table 1 summarizes their main purposes, typical scales, and intended users in their original settings.

Table 1: Summary of the nitrogen footprint, nitrogen use efficiency, and national/regional nitrogen budgets

| Approach                           | Main purpose                                                                                                                           | Typical scales                                 | Intended users                                         |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|--------------------------------------------------------|
| Nitrogen footprint                 | Link many kinds of entities of supply chains, primarily consumers, to nitrogen emissions and impacts                                   | Individual, institutional, watershed, national | Individual, institutional, watershed, national, global |
| Nitrogen use efficiency (NUE)      | Assess how efficiently nitrogen inputs are converted into useful outputs within a defined system and what it means at different scales | Farm, food system; also regional to global     | Farmers, other producers, advisors, and policy makers  |
| National/regional nitrogen budgets | Understanding N cycle, identifying storage, losses, and possible intervention points                                                   | Subnational region, country, and global scale  | Policy makers, researchers                             |

Based on Table 3.1 in [the INMS Guidance Document on Nitrogen Impact Assessment Methods](#).

Note: Here, “nitrogen footprint” mainly refers to the N-Calculator and related derived approaches, while some other footprint approaches, including a budget-based approach, may also be within scope.